



Homework 4: Model-Free RL Temporal Difference and Exploration

Environment Description

The agent no longer knows the transition probabilities P . It must learn from experience. We simplify the battery states to S_L (Low) and S_H (High). The actions are a_S (Save) and a_B (Boost).

Assume the agent experiences the following trajectory (state, action, cost, state):

$$(S_L, a_S, 10, S_L) \rightarrow (S_L, a_B, 20, S_H) \rightarrow (S_H, a_S, 0, S_H)$$

1. SARSA vs. Q-Learning

1. Write the update rule for Q-Learning for the third transition: $(S_H, a_S, 0, S_H)$.
2. Assume that the agent's policy decides to explore and chooses a_B as the action for the fourth transition. Would the update of the third transition for SARSA differ from Q-Learning? Explain.
3. Are these methods *bootstrapping*? Explain.

2. Exploration-Exploitation

The agent uses ϵ -greedy with $\epsilon = 0.2$.

1. If $Q(S_H, a_S) = 5$ and $Q(S_H, a_B) = 15$, what is the probability of choosing a_B ?
2. What is the risk of decreasing ϵ to 0 (pure exploitation) too quickly?

3. Model-Based Learning (MLE)

If the agent ignores TD learning and instead tries to reconstruct the environment model (P matrix) from the trajectory provided in the Description:

1. Based **only** on that specific trajectory, what is the Maximum Likelihood Estimate (MLE) for $P(S_L|S_L, a_S)$ and $P(S_H|S_L, a_S)$? Show the transition count table used to derive these probabilities.

4. Policy Improvement Step

Assume after several steps the agent has learned: $Q(S_L, a_S) = 40$, $Q(S_L, a_B) = 30$, $Q(S_H, a_S) = 10$, $Q(S_H, a_B) = 5$.

1. Demonstrate the result of a Greedy Policy Improvement step. Write down the new deterministic policy $\pi(s)$ for both states.
2. If the agent was using a Softmax policy with temperature $\tau = 10$ instead of ϵ -greedy, show the calculation for the probability of choosing a_S in state S_L .